SEPM LAB PRACTICAL 8

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Aim: To study white-box testing and path testing, draw a Control Flow Graph (CFG) for a program, calculate cyclomatic complexity, identify independent paths, and prepare test cases.

What is White-Box Testing:
White-box testing is a software testing technique in which the internal structure, source code, statements, conditions, and control flow of a program are examined. The tester uses knowledge of the program code to design test cases.
Objectives:
- To verify the internal logic of a program.
- To identify logical errors.
- To check statements, branches, and paths.
- To improve test coverage.

What is Path Testing: Path testing is a white-box testing technique in which test cases are designed to execute selected paths through a program's Control Flow Graph (CFG). A path is a sequence of nodes and edges followed during program execution. Path testing helps verify different possible execution routes through the program.

Program:
Write a Python program that accepts a student's marks and checks whether the marks are valid. Marks outside the range 0 to 100 are invalid. For valid marks, the program displays Pass if marks are 40 or above; otherwise, it displays Fail.

```python
marks = int(input("Enter student's marks: "))

if marks < 0 or marks > 100:
    print("Invalid marks")
elif marks >= 40:
    print("Pass")
else:
    print("Fail")
```

Control Flow Graph (CFG):

A Control Flow Graph represents the flow of execution in a program. Nodes represent statements or conditions, and directed edges represent the flow between them.

Nodes:

1 – Start

2 – Input marks

3 – Check marks < 0 or marks > 100

4 – Print Invalid marks

5 – Check marks >= 40

6 – Print Pass

7 – Print Fail

8 – End

Edges:

1 → 2

2 → 3

3 (True) → 4

3 (False) → 5

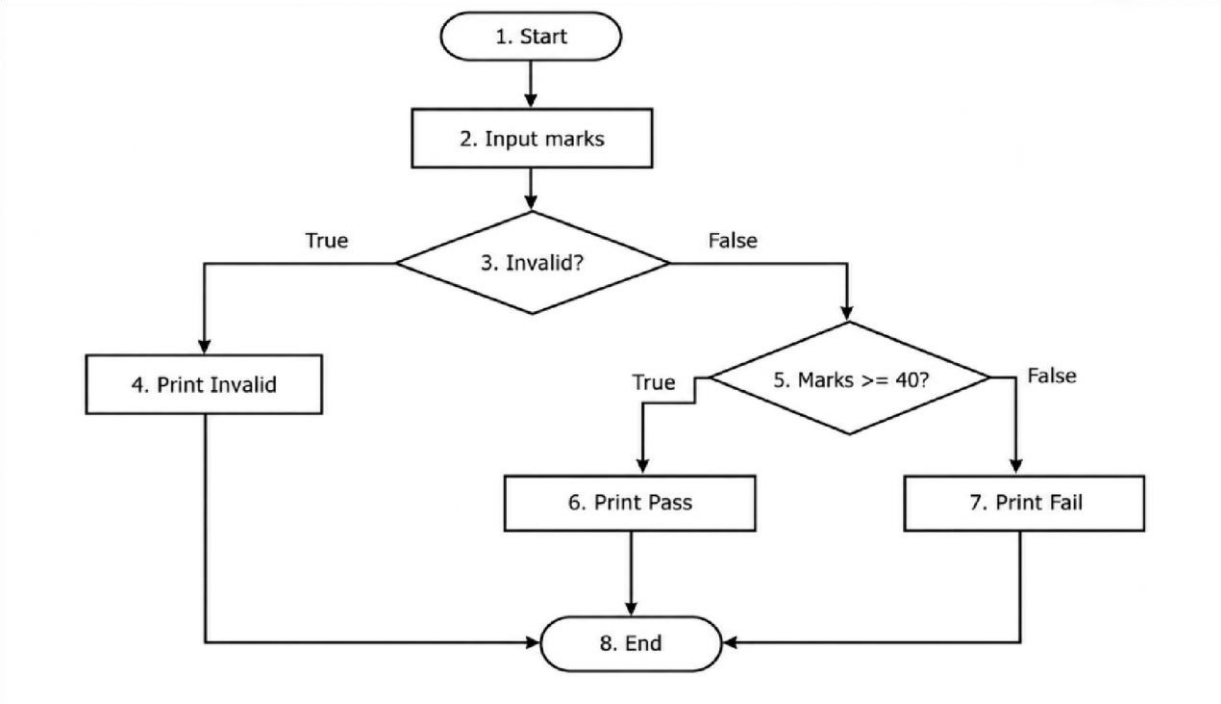
4 → 8

5 (True) → 6

5 (False) → 7

6 → 8

7 → 8



Cyclomatic Complexity:

Cyclomatic complexity gives the number of linearly independent paths in a program.

Formula: $V(G) = E - N + 2P$

Where E = number of edges, N = number of nodes, and P = number of connected components.

For this program:

$E = 9, N = 8, P = 1$

$V(G) = 9 - 8 + 2(1) = 3$

There are 2 decision points, so $V(G) = 2 + 1 = 3$.

Therefore, the cyclomatic complexity is 3.

Independent Paths:

Since the cyclomatic complexity is 3, three independent paths are identified:

Path 1 – Invalid marks: 1 → 2 → 3 (True) → 4 → 8

Path 2 – Pass: 1 → 2 → 3 (False) → 5 (True) → 6 → 8

Path 3 – Fail: 1 → 2 → 3 (False) → 5 (False) → 7 → 8

Test Cases:

Test Case ID	Input Marks	Expected Output	Path
TC-01	-5	Invalid marks	Path 1
TC-02	75	Pass	Path 2
TC-03	28	Fail	Path 3